Designing Microsoft Azure Infrastructure Solutions: AZ:305

TECHNOLOGY

Design a Solution for Logging and Monitoring

Learning Objectives

By the end of this lesson, you will be able to:

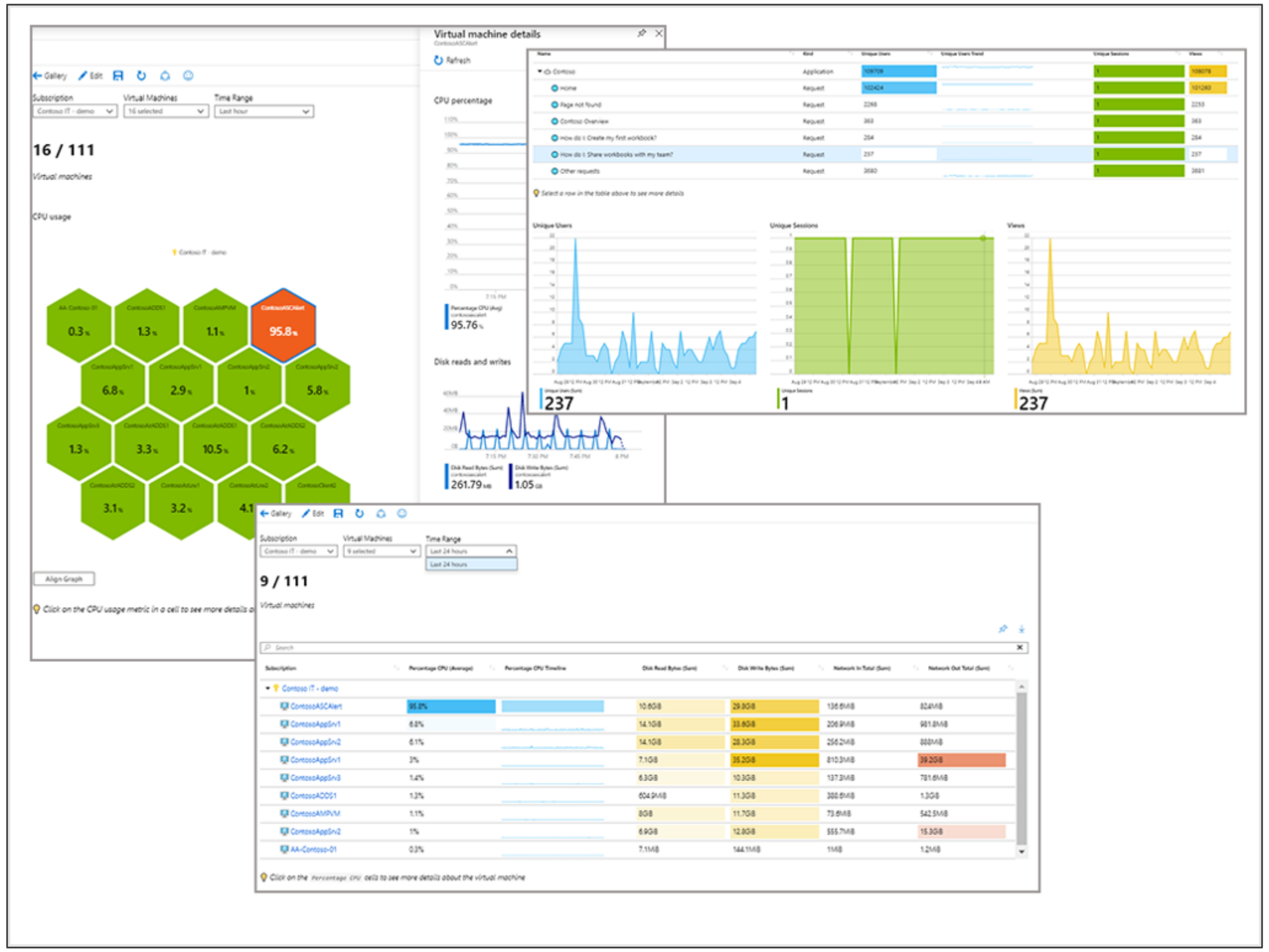
- Implement appropriate monitoring tools for maintaining the health, performance, and security of Azure resources
- Analyze Azure Monitor health and availability monitoring to proactively identify and address issues
- Initiate automated responses using action groups for proactive issue resolution, minimizing downtime, optimizing resource utilization
- Configure and manage alerts for important notifications

TECHNOLOGY

Design for Azure Workbooks and Azure Insights

What Are Azure Workbooks?

Azure Workbooks provide a dynamic platform for the processing of data and the development of effective visual reports.



It also allows users to access data from various Azure datasets and merge them to create a seamless interactive experience.

Source: https://learn.microsoft.com/en-us/azure/azure-monitor/visualize/workbooks-overview

Application Insights

Application Insights is a feature of Azure Monitor. It is a developer and DevOps-focused application performance management service.

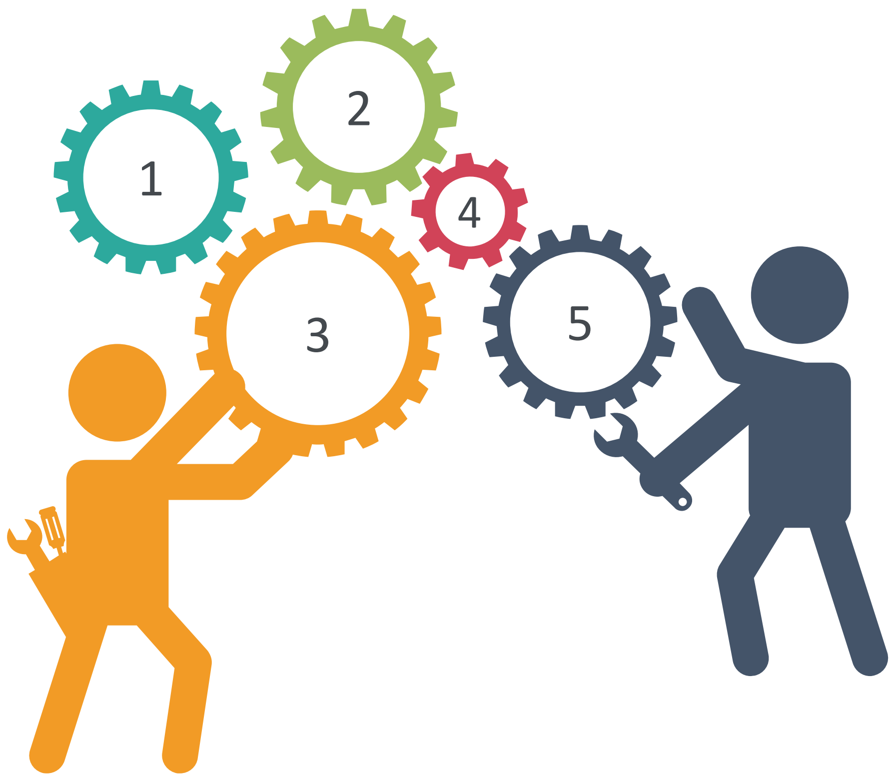
Azure Monitor VM

Azure Monitor is a full-featured monitoring service that lets the user track all the Azure resources.

Azure Monitor's functions are linked to the Azure site for the Azure services it monitors. Users do not need to interact with it directly to do a range of monitoring tasks.

Azure Container Insights

Container insights is a feature that allows the user to track the effectiveness of container workloads on the following platforms:

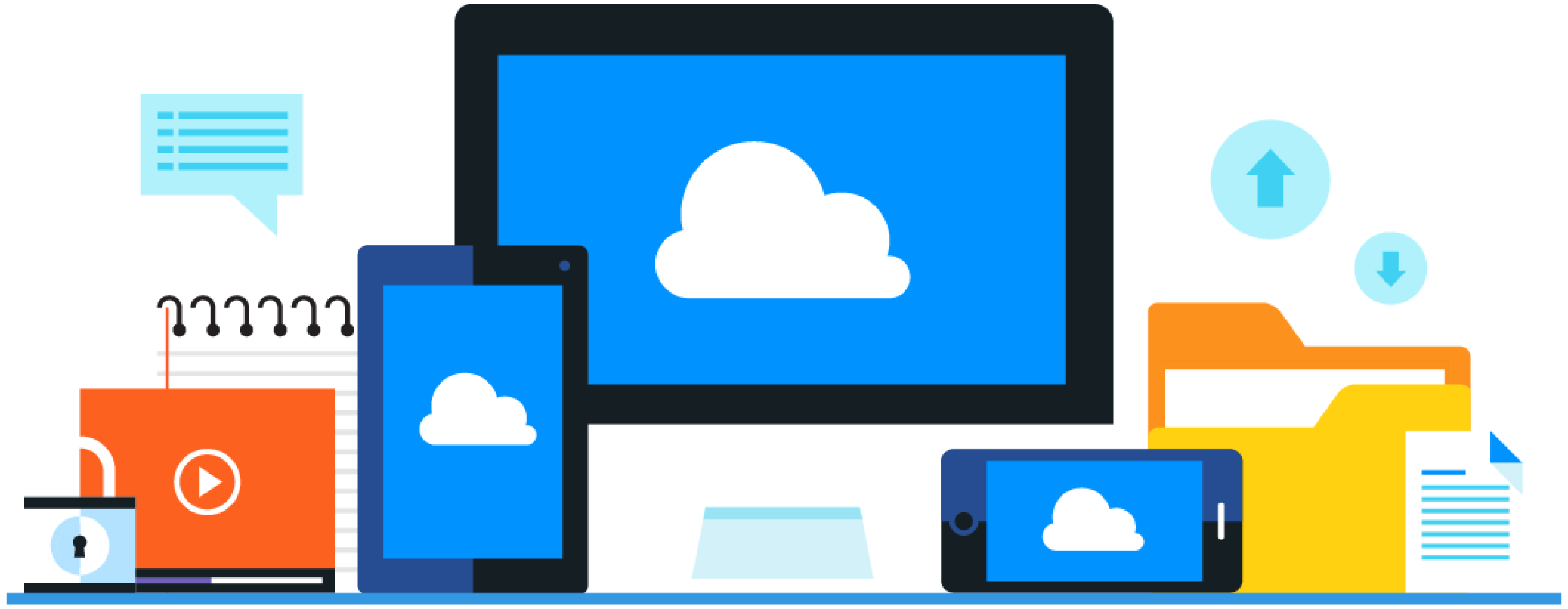


- Managed Kubernetes clusters hosted on Azure Kubernetes Service (AKS)
- Self-managed Kubernetes clusters hosted on Azure using AKS Engine
- Container Instances in Azure
- Self-managed Kubernetes clusters hosted on Azure Stack
- Azure Arc-enabled Kubernetes

Recommend Appropriate Monitoring Tools for a Solution

Azure Monitoring

It collects and analyzes data to assess the application's performance, health, availability, and the resources that it depends on.



Azure Infrastructure Monitoring

The following are the different ways of how Azure monitoring works:

- Configuration and change management
- Vulnerability management
- Vulnerability scanning
- Protective monitoring
- Incident management

Azure reviews and updates configuration settings and baseline configurations of hardware, software, and network devices annually.

Azure Infrastructure Monitoring

The following are the different ways of how Azure monitoring works:

- Configuration and change management
- Vulnerability management
- Vulnerability scanning
- Protective monitoring
- Incident management

Security update management helps protect systems from known vulnerabilities.

Azure Infrastructure Monitoring

The following are the different ways of how Azure monitoring works:

- Configuration and change management
- Vulnerability management
- Vulnerability scanning
- Protective monitoring
- Incident management

It is performed on server operating systems, databases, and network devices.

Azure Infrastructure Monitoring

The following are the different ways of how Azure monitoring works:

- Configuration and change management
- Vulnerability management
- Vulnerability scanning
- Protective monitoring
- Incident management

Monitoring tools like Microsoft Monitoring Agent (MMA) and System Center Operations Manager are used for active monitoring.

Azure Infrastructure Monitoring

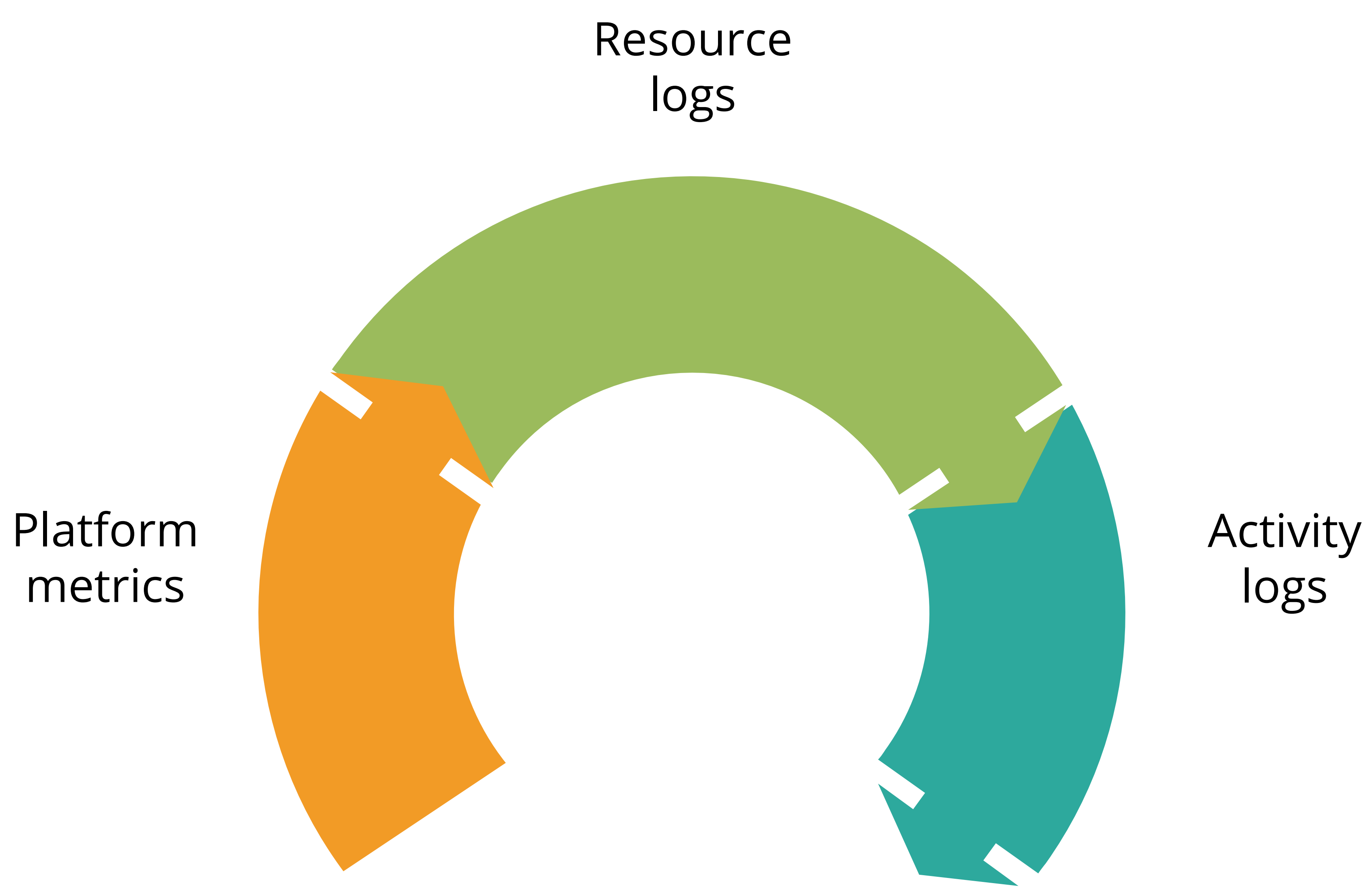
The following are the different ways of how Azure monitoring works:

- Configuration and change management
- Vulnerability management
- Vulnerability scanning
- Protective monitoring
- **Incident management**

Microsoft implements a security incident management process to facilitate a coordinated response to incidents.

Monitoring Data

The types of monitoring data are as follows:



Monitoring Data

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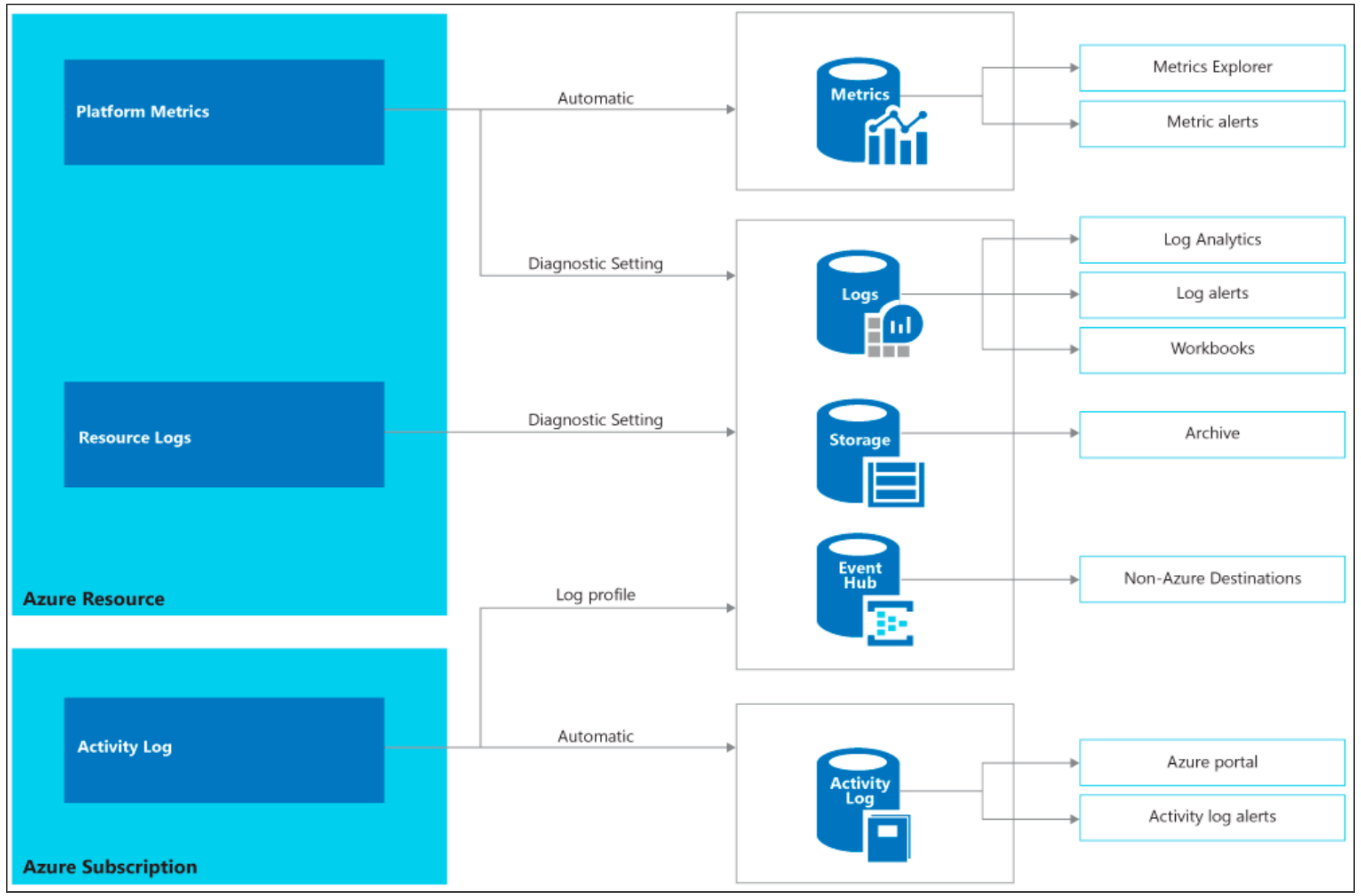

Platform metrics

Provides diagnostic and numerical values that are automatically collected at regular intervals, which describe some aspects of a resource at a time

Monitoring Data

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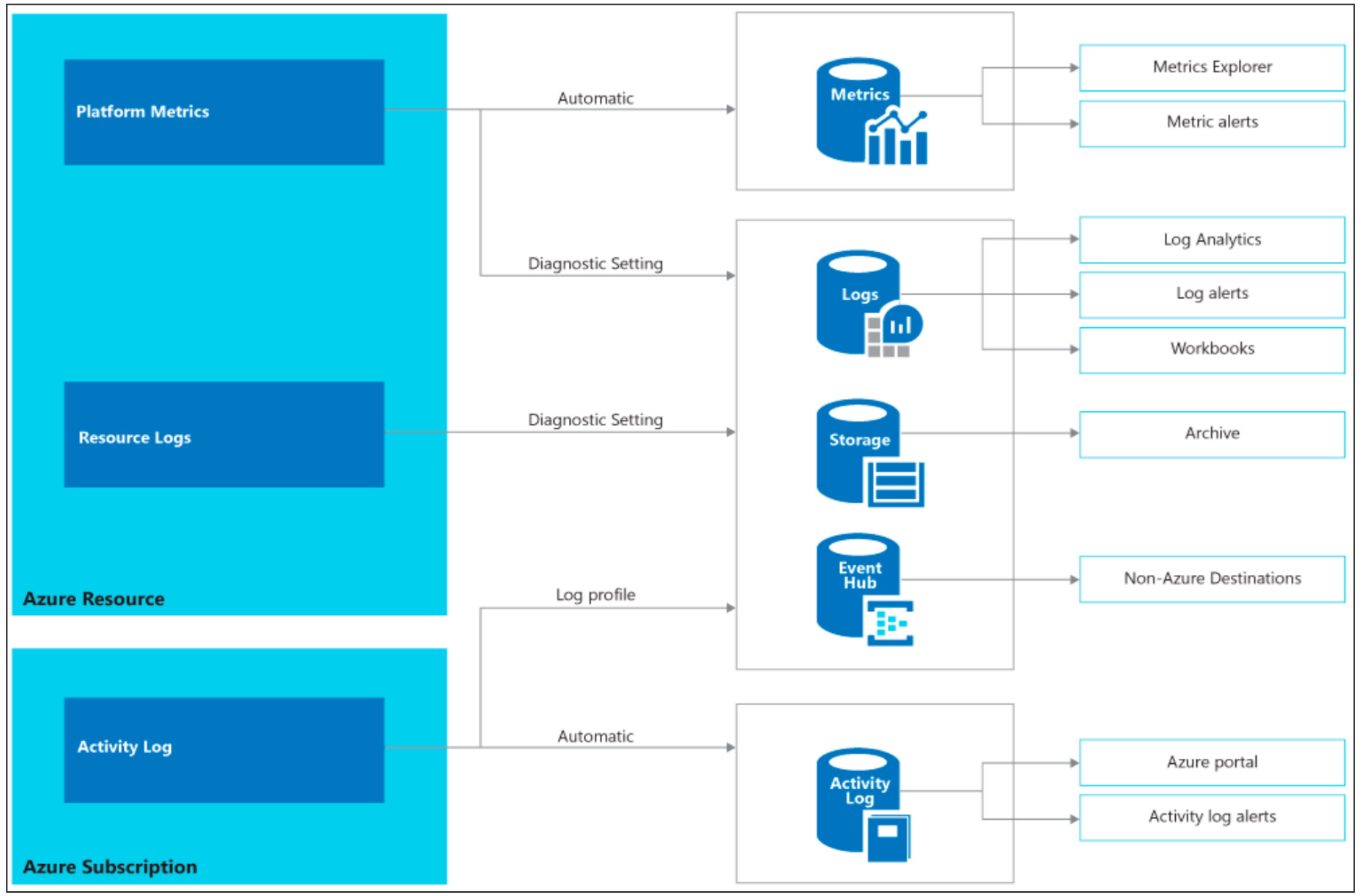
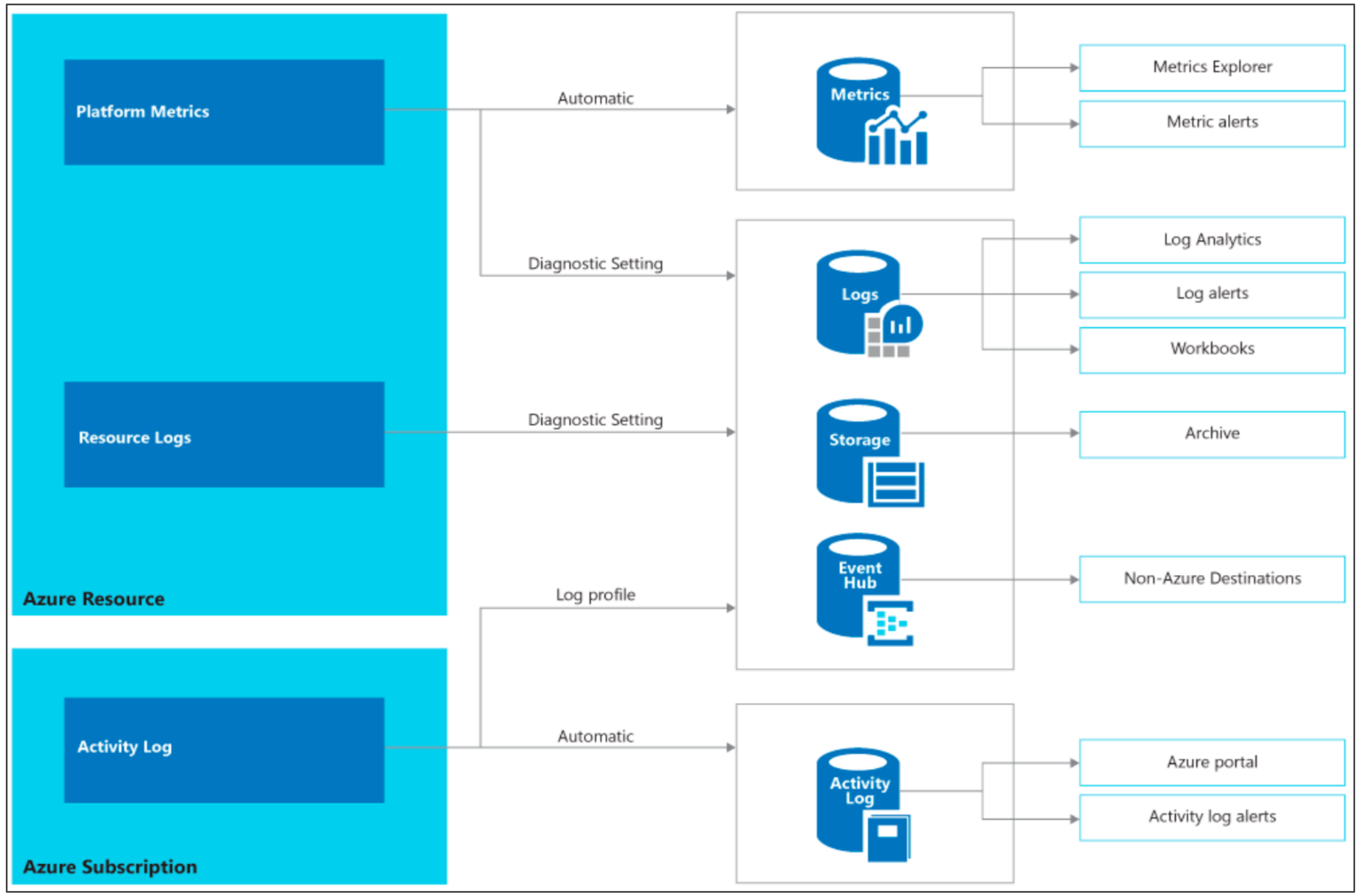


Resource logs

Provides insights into operations that were performed within an Azure resource (the data plane)

Monitoring Data

The types of monitoring data are as follows:

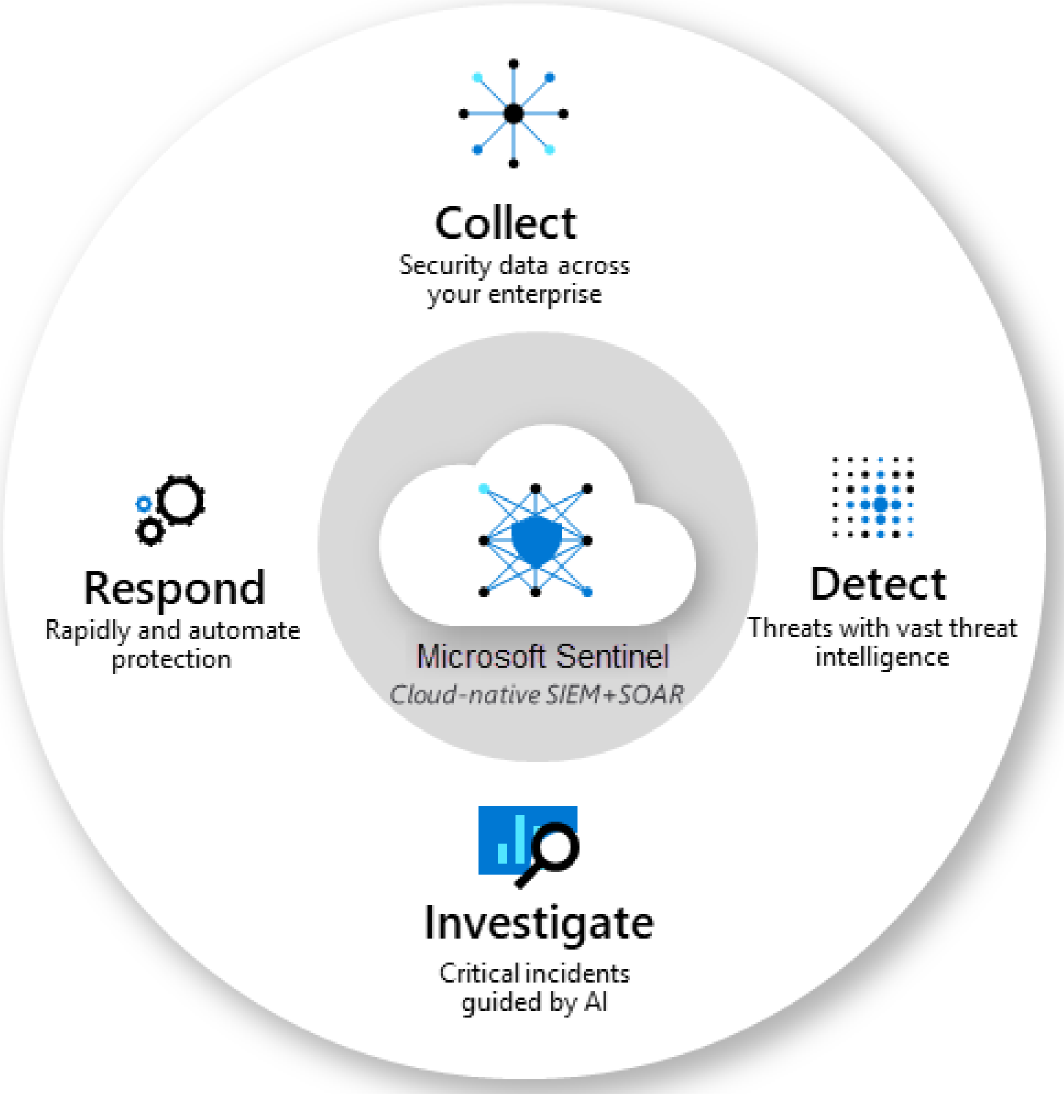


Activity log

Provides insight into the operations on each Azure resource in the subscription from the outside (the management plane)

Microsoft Sentinel

It is a built-in threat intelligence for detection and investigation.



- Collects data from devices, users, infrastructure, and applications
- Helps with cloud and on-premises monitoring and management
- Investigates threats using AI

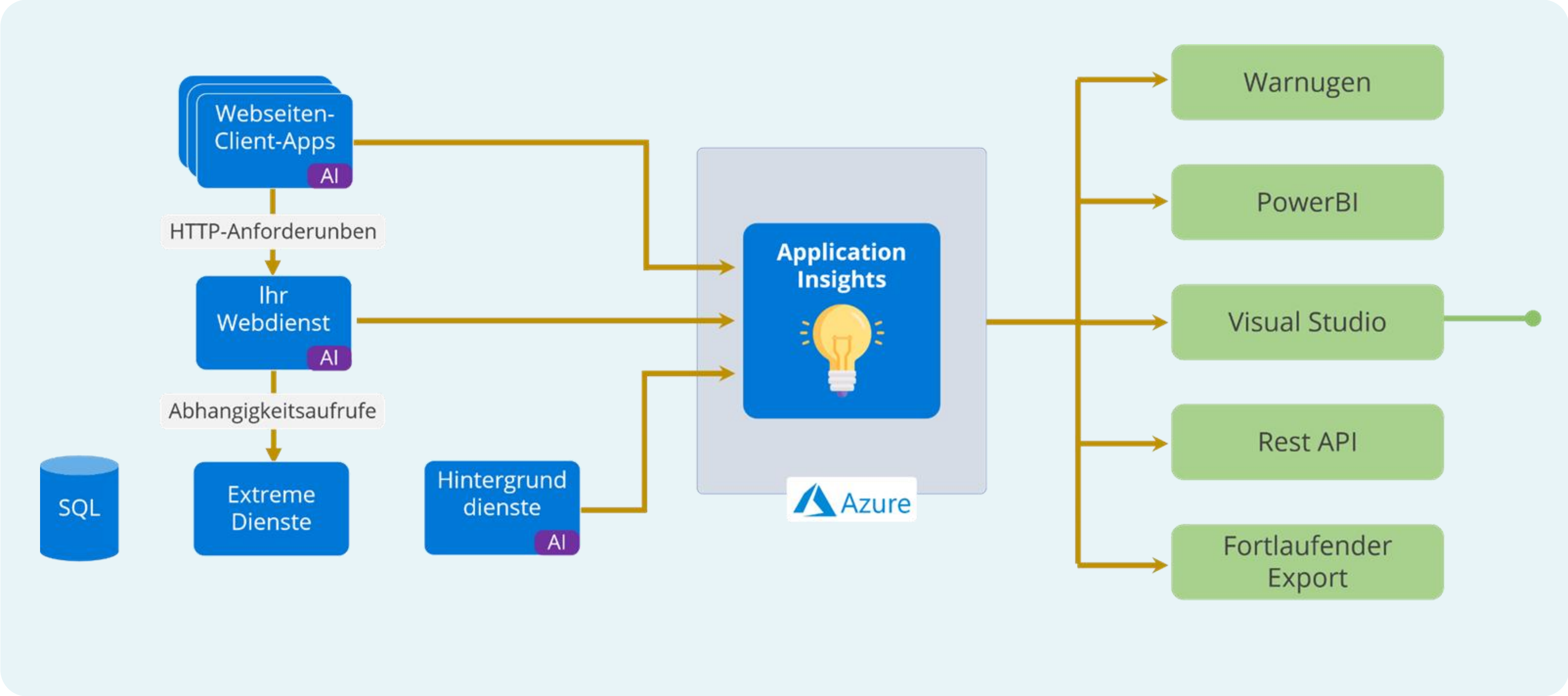
Application Monitoring

Connection points to a range of development tools are available in application insights.

Application insights keep track of user’s web application availability, performance, and usage.

Application Monitoring

These are the different application insights:



- Default dashboard with most important metrics
- Smart detection
- Usage analysis
- Snapshot debugger
- Performance statistics from a client and server

Platform Monitoring

These are the container insights:

- Clusters, nodes, and pods are visualized, and actionable information is provided.
- CPU, memory, and logs for individual Kubernetes pods are also available.
- Container logs are also collected.

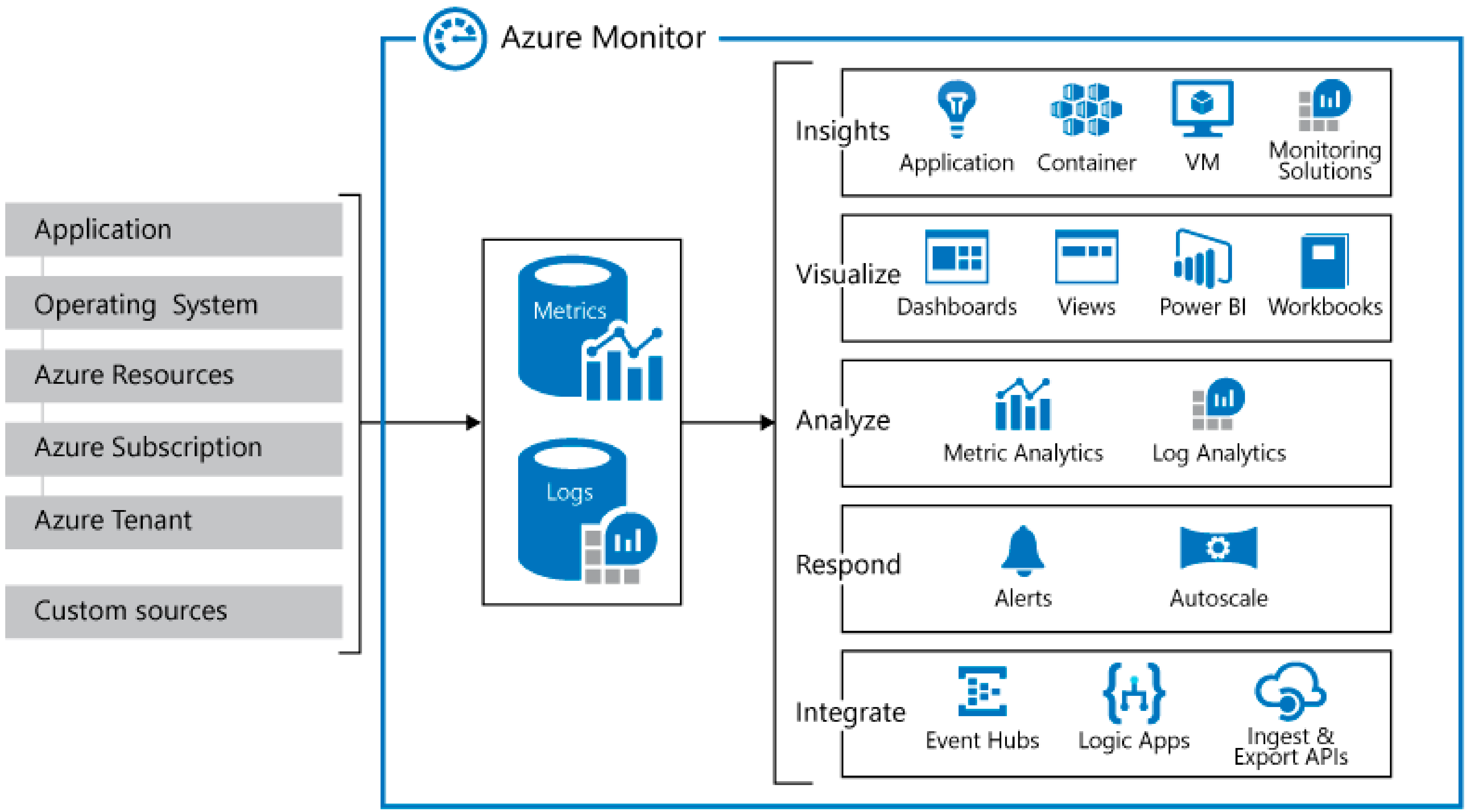
Monitoring: Best Practices

These are the best practices for monitoring:

- Ensure applications are performing as expected
- Ensure applications are as reliable as their underlying infrastructure
- Ensure quality through continuous deployment
- Prepare role-based dashboards and workbooks

Azure Monitor Service

Azure includes multiple services that perform a specific role or task in the monitoring space.



Azure Monitoring: Features

These are the key features used for Azure monitoring:

Monitor and visualize metrics

Metrics are numerical values available from Azure Resources that help the user understand the health, operation, and performance of the system.

Azure Monitoring: Features

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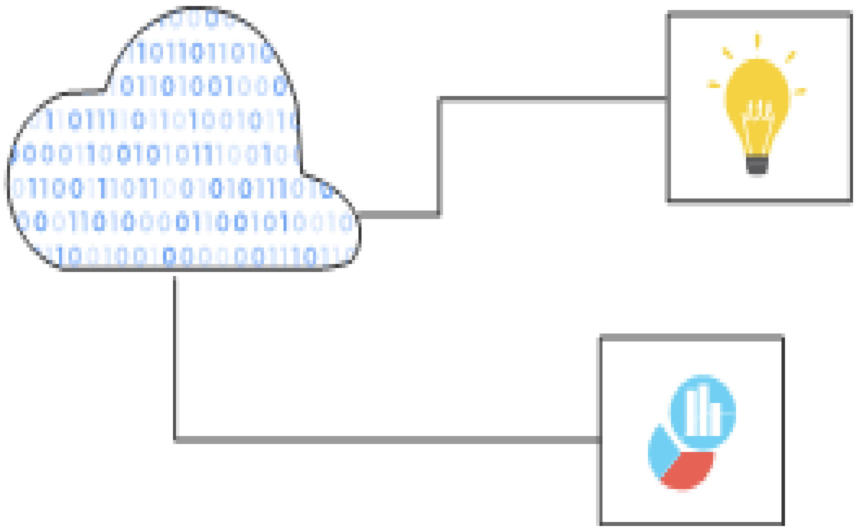


Query and analyze logs

Logs are activity logs, diagnostic logs, and telemetry from monitoring solutions. Analytics queries help with troubleshooting and visualization.

Azure Monitoring: Features

These are the key features used for Azure monitoring:

Setup alerts and actions

Alerts notify critical conditions and take corrective automated actions based on triggers from metrics or logs.

Metrics

They are numerical values that describe an aspect of a system at a point in time.

They are lightweight and capable of supporting real-time scenarios.

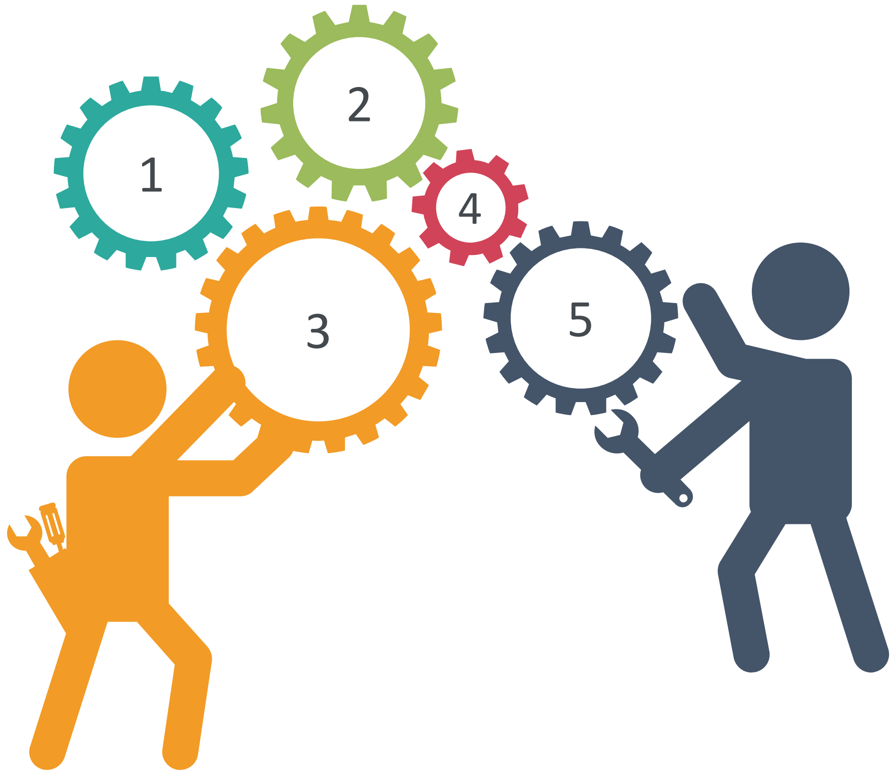
Log Data

Logs contain different types of data organized into records with different sets of properties for each type.

Telemetries, such as events and traces, are stored as logs in addition to performance data so that they can be combined for analysis.

Azure Monitor Data Types

Azure Monitor collects data from each of the following tiers:



Azure subscription monitoring data

Application monitoring data

Guest OS monitoring data

Azure resource monitoring data

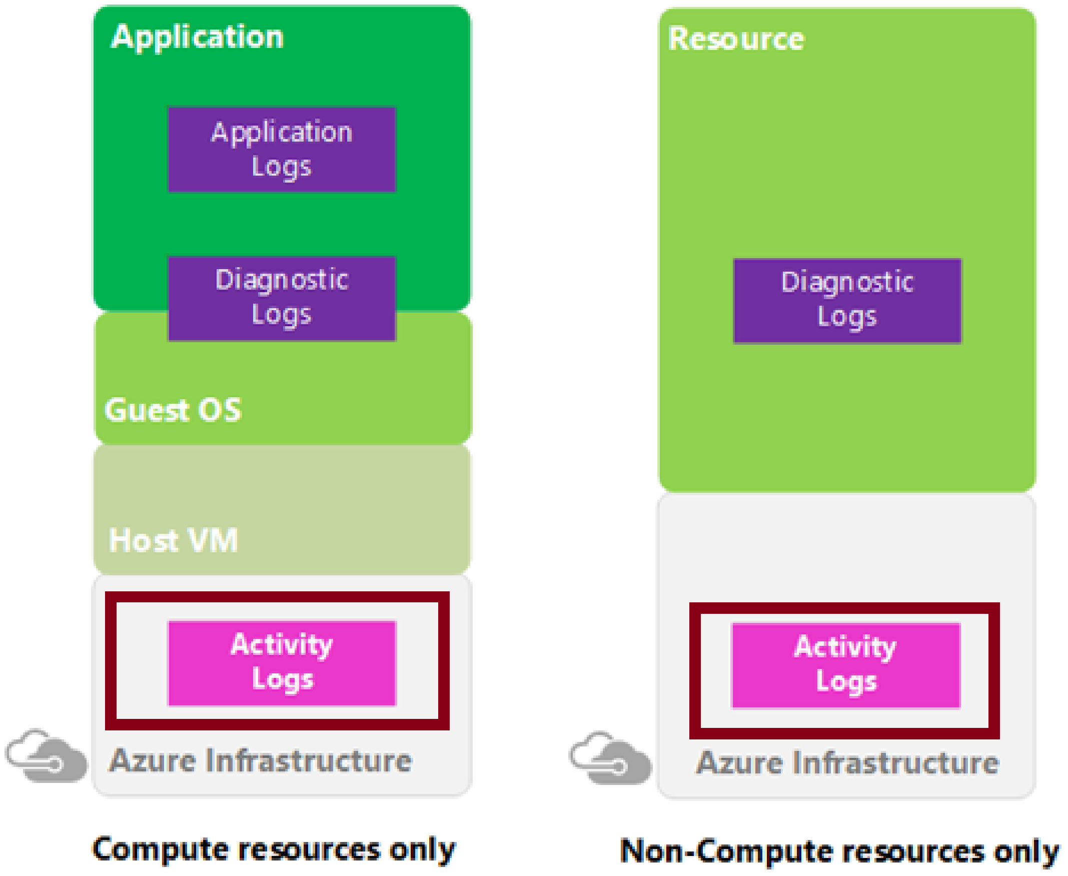
Azure tenant monitoring data

Azure Advisor

Azure Advisor is a personalized cloud consultant that helps follow best practices by optimizing Azure deployments.

Activity Log

It is a subscription log that provides insight into subscription-level events that have occurred in Azure.



Activity log can determine:

- What operations (PUT, POST, DELETE) were performed on all resources?
- Who started the operation?
- When did the operation occur?
- What is the status of the operation?
- What are the values of other properties that might help in researching the operation?

View the Activity Log

In the Azure portal, users can filter the Activity log by:

Subscription	Resource group	Event initiated by
Timespan	Resource name	Operation name
Event severity	Resource type	Search

App Service Diagnostic Logs

The Azure App Service has built-in diagnostics that can help users debug apps.

This service is not enabled by default.

Diagnostic Logs Types

There are two categories of Azure Web App diagnostic logs:

Application diagnostic logs:

Contains information produced by the application code

Web Server diagnostic logs:

Contains information produced by the web server that the web application is running on

Diagnostic Logs Types

There are three types of web server diagnostic logs that user can enable:

Web Server Logging

- Contains all HTTP events on a website and is formatted using the W3C extended log file format

Detailed Error Messages

- Contains information on requests that resulted in an HTTP status code of 400 or higher

Failed Request Tracing

- Contains detailed traces of any failed requests
- Contains traces for all the IIS components that were involved in processing the request

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Log Structure : Log File Type and Location

Web server diagnostic logs are stored in different directories as shown below:

Web server logs	D:\Home\LogFiles\http\RawLogs\
Detailed error logs	D:\Home\LogFiles\DetailedErrors\
Failed request logs	D:\Home\LogFiles\LogFiles\W3SVC####\

TECHNOLOGY

Health and Availability Monitoring

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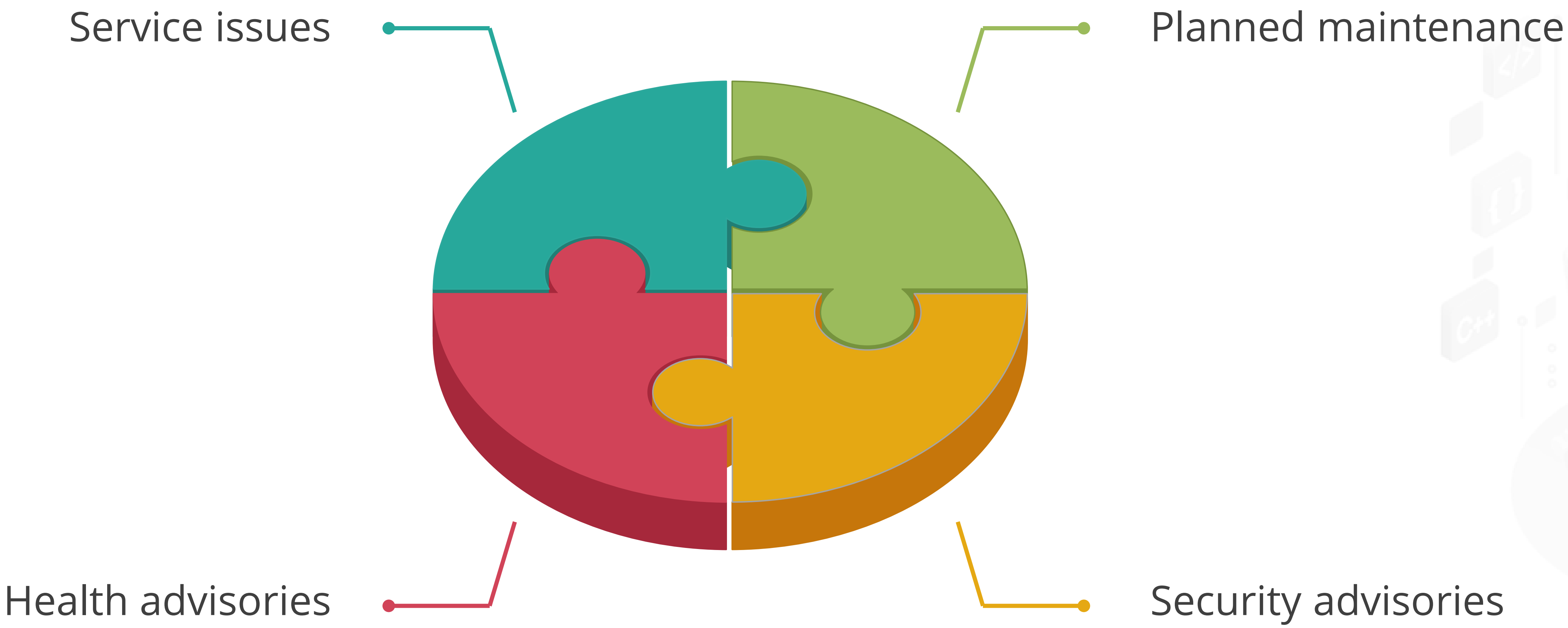
Azure Service Health

- The Microsoft Azure platform is transparent in terms of service availability.
- Service Health Blade provides a global map view of regions with service issues, reviews planned maintenance, is informed of health advisories, and looks at the health history.

Azure Service Health

Service health events are as follows:



Azure Service Health

Service health allows the user to:

- View the current issues that impact the services
- Fetch links and downloadable explanation
- Get support from Microsoft
- Pin a personalized health map to the dashboard
- Configure service health alerts

Azure Resource Health

It helps in the diagnosis and resolution of service issues affecting Azure resources.

It provides:

- Resource definition and health assessment
- Health status
  - Available
  - Unavailable (includes platform events and non-platform events)
  - Unknown
  - Degraded
- Reporting of an incorrect status
- History information

Cost Monitoring

Monitoring Azure Costs

Cost monitoring is about establishing controls and business processes for reviewing cloud spend to avoid any misuse and take advantage of new opportunities through the flexibility provided by cloud.

Cost Optimization

Consider these trade-offs between cost optimization and other aspects of design, such as security, scalability, resilience, and operability. An optimal design is not synonymous with a low-cost design.

Cost versus reliability

- Does the cost of high-availability components exceed the acceptable downtime?

Cost versus performance efficiency

Factors impacting performance:

- Fixed or consumption-based provisioning
- Azure regions
- Caching
- Batch or real-time processing

Cost Optimization

Cost versus security

Increasing the security of the workload will increase the cost.

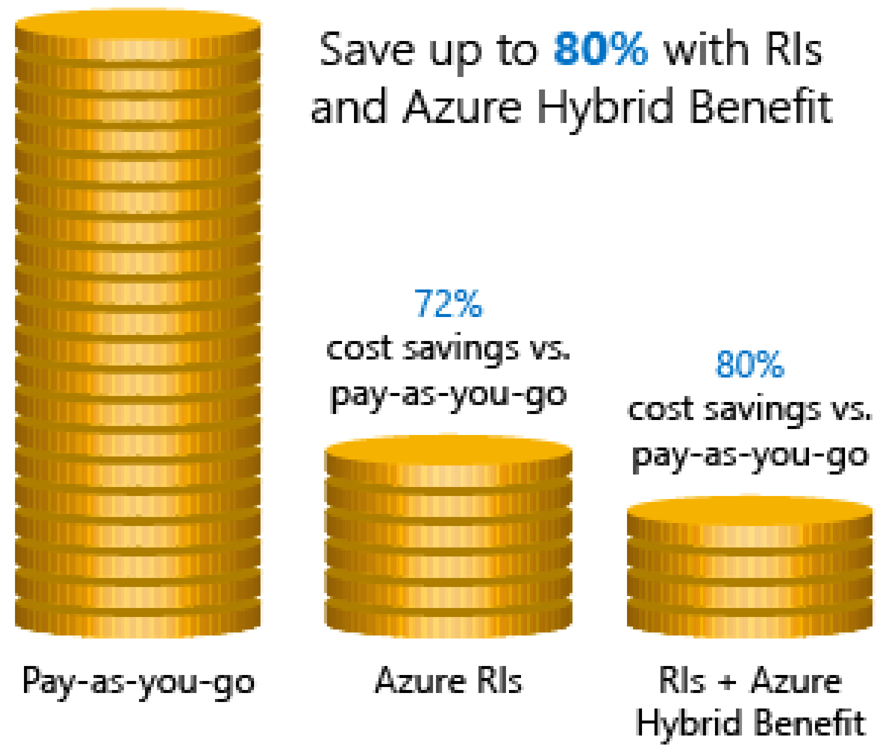
For example, for specific security and compliance requirements, deploying to differentiated regions will be more expensive. Premium security features can also increase the cost.

Cost versus operation excellence

It might increase the cost initially, but it can reduce the cost over time.

For example, some patterns might introduce flows that span multiple components or shift processes from the server to the client. These changes can increase the complexity of correlating and tracking information.

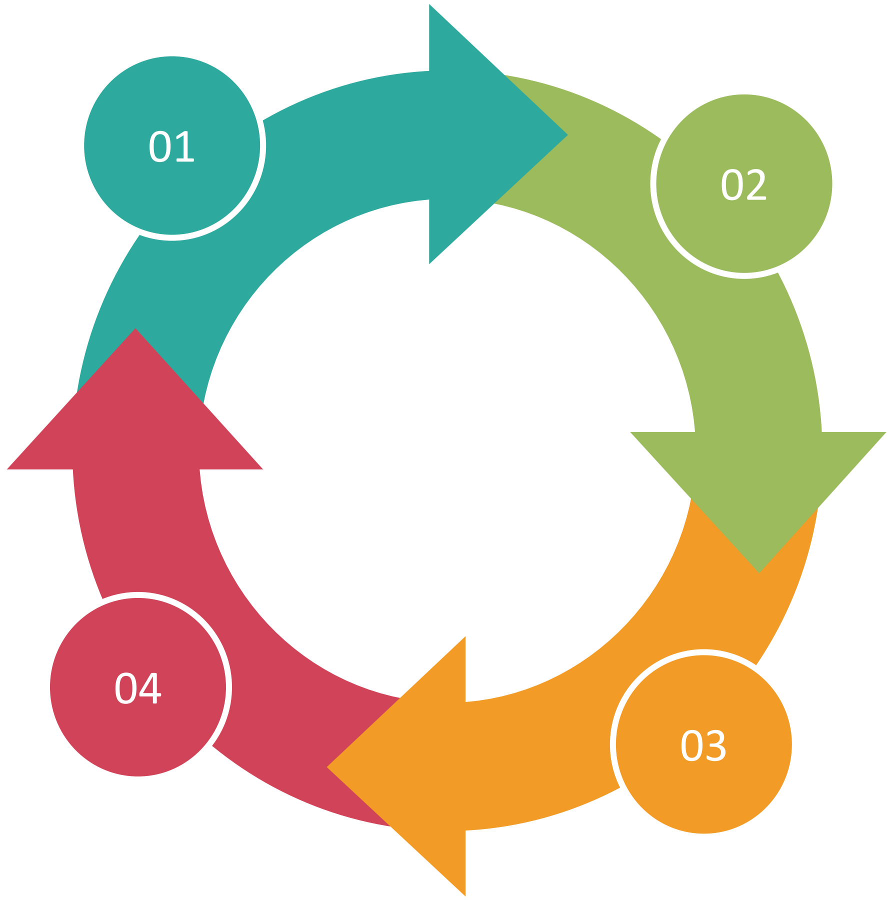
Cost Savings



- **Azure Reservations:** These help users save money by allowing them to pay for services in advance.
- **Azure Hybrid Benefits:** With Software Assurance, a user can use on-premise licenses for Windows Server and SQL Server.
- **Azure Credits:** These are the benefits of a monthly credit that allows a user to try, build, and test new Azure solutions.
- **Regions:** A user must choose low-cost locations and regions.

Report on Spend

Azure provides tools to create cost reports:



Azure Advisor Cost Analysis tool

Provides an overview of spending over a period to understand the spending trend

Azure Advisor recommendations

Highlights over-provisioned services and recommends ways to lower costs

Power BI Desktop

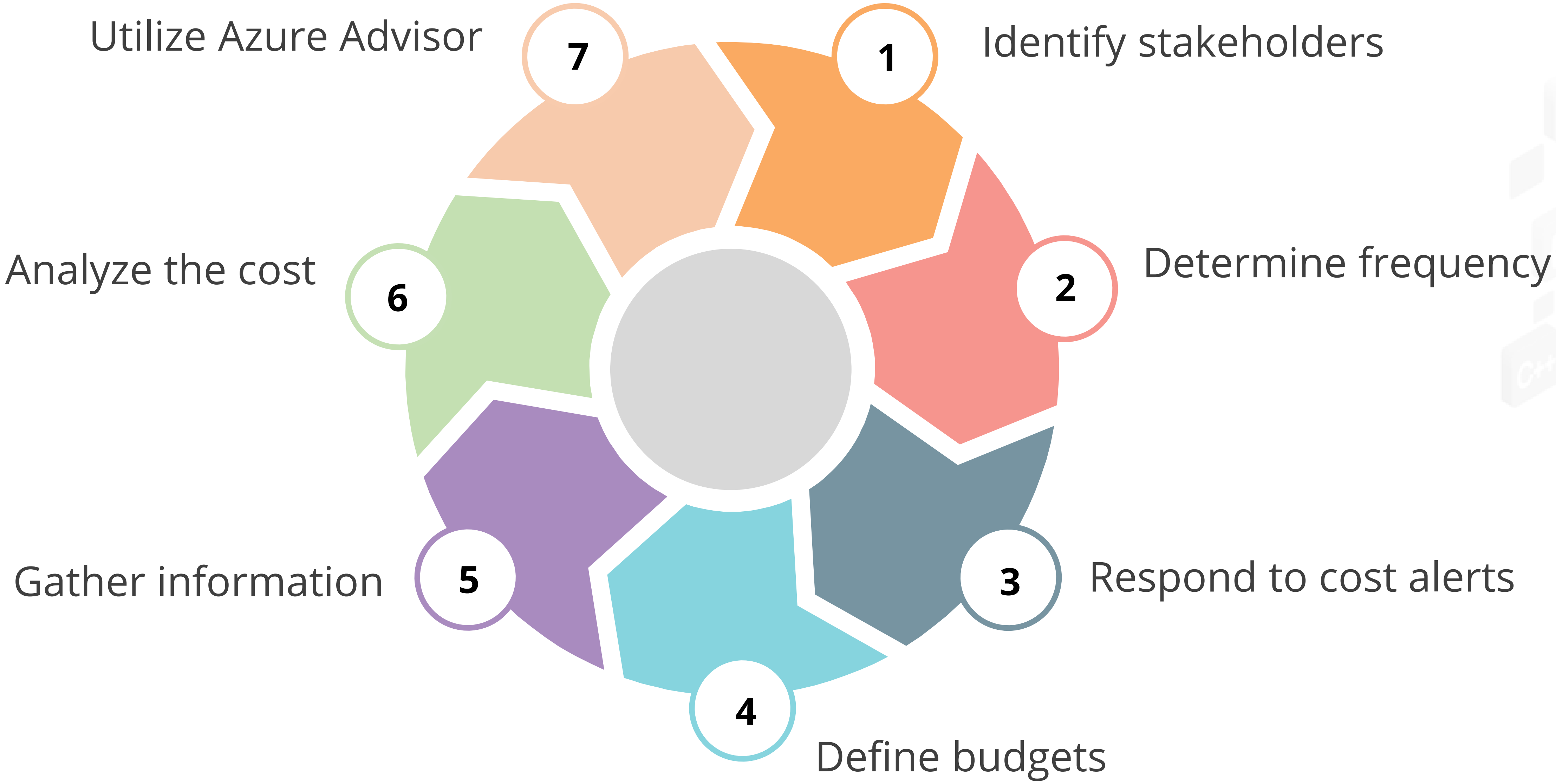
Connects to billing data from Azure Cost Management

Consumption APIs

Are provided by Azure, so a user can write custom tools and scripts to track costs over time

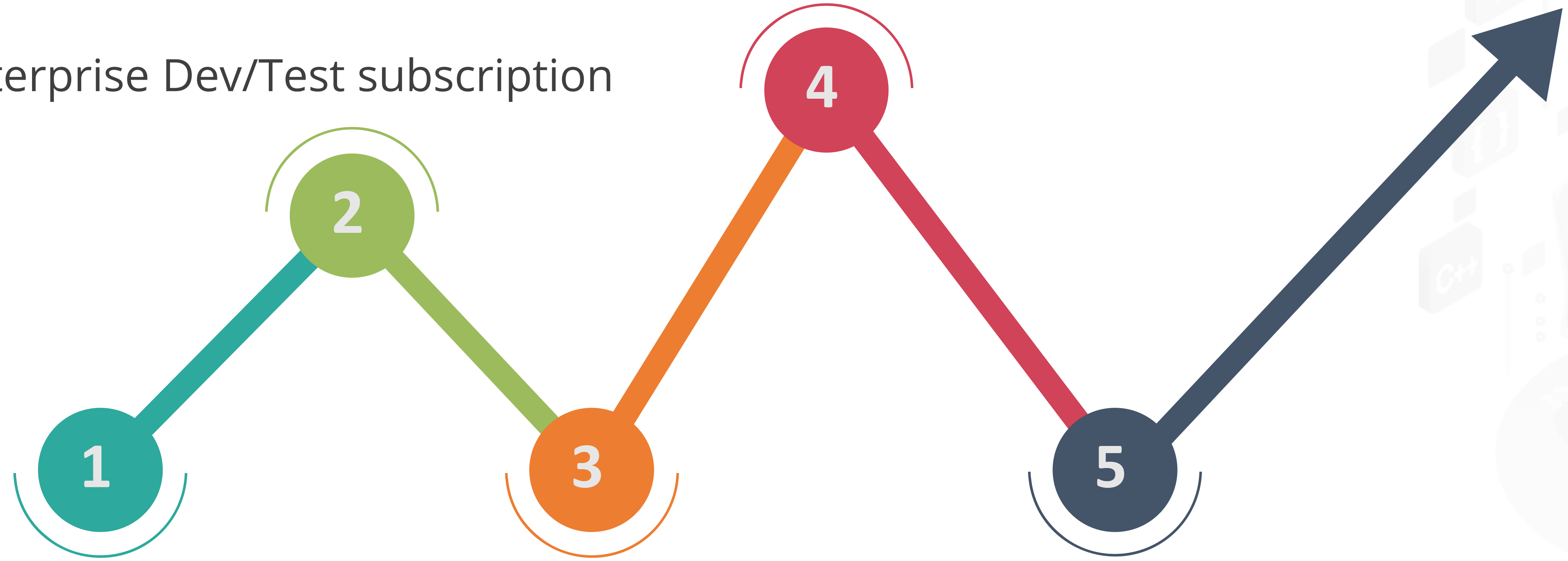
Monitoring Azure Costs

To monitor Azure cost, you need to:



Optimizing Azure Costs

Consider the following methods of managing pricing:



1. Enterprise agreement
2. Enterprise Dev/Test subscription
3. Cloud service provider (partner program)
4. Azure Hybrid Benefit
5. Azure Reserved Instances

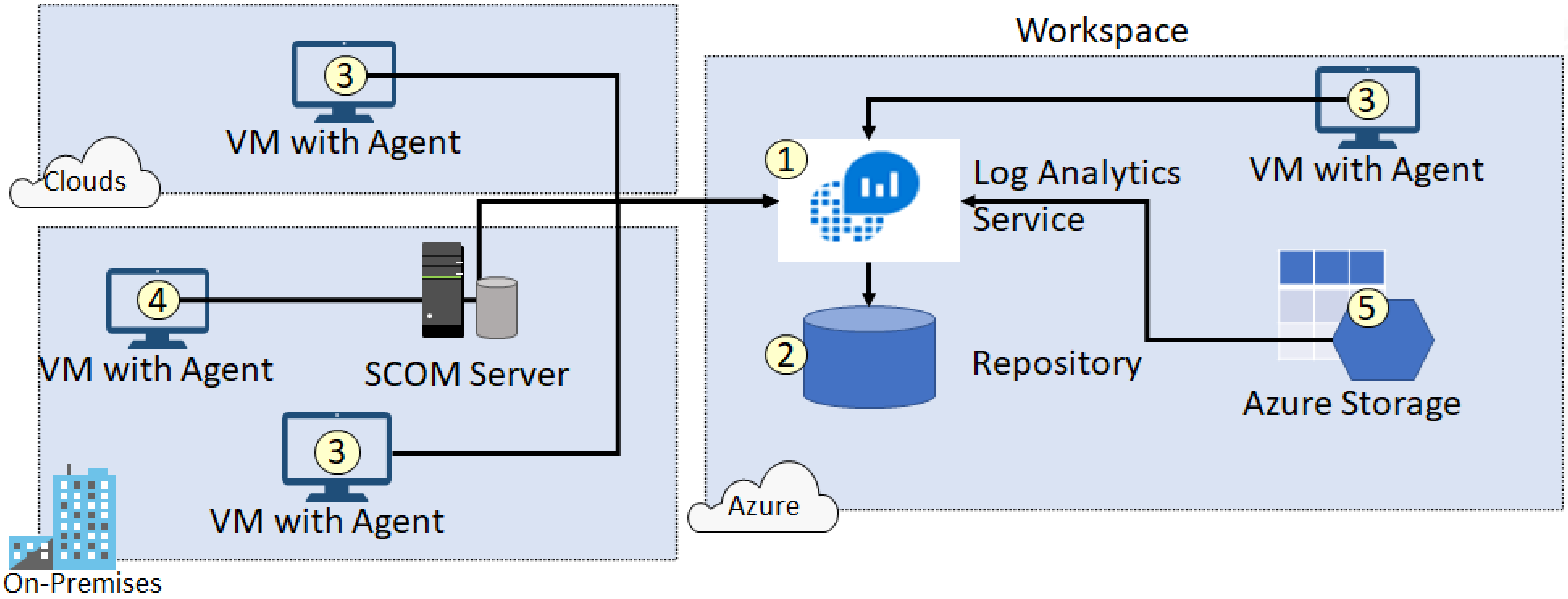
Advanced Logging

Log Analytics

Log analytics collect and analyze data generated in the cloud and on-premise systems.

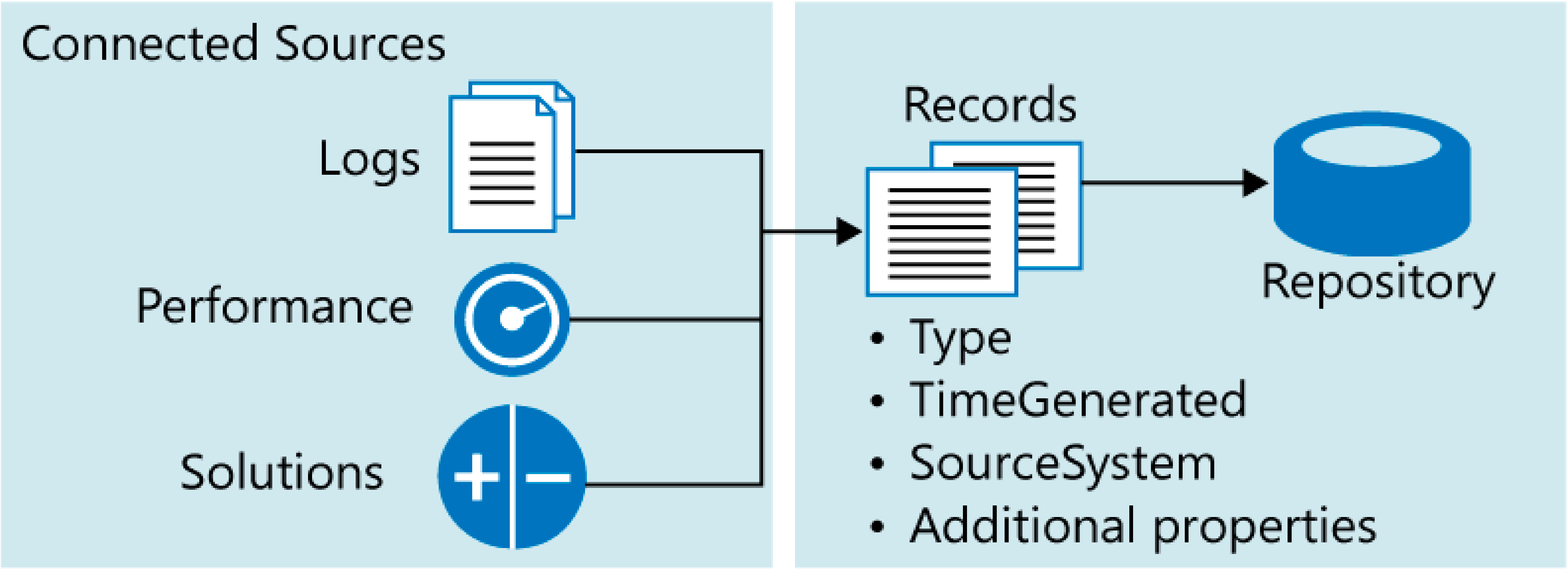
Connected Sources

Connected sources are the computers and other resources that generate data collected by Log Analytics.



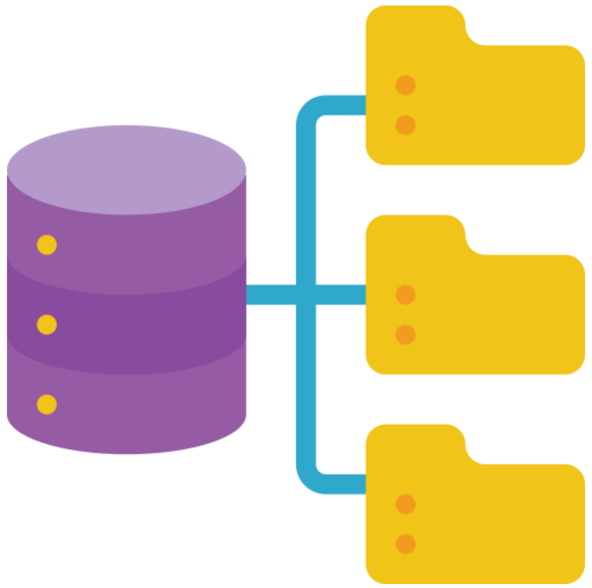
Data Sources

Data sources represent the data collected from connected sources.



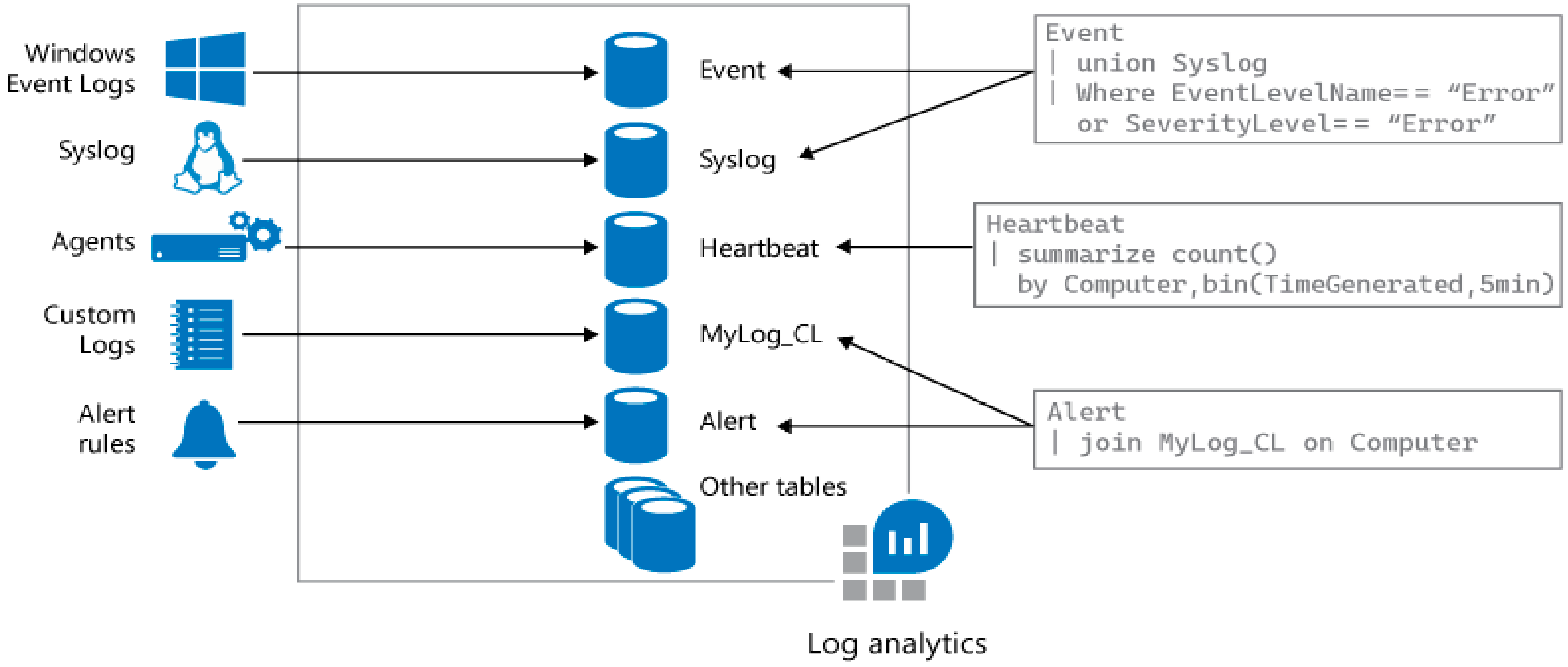
Log Analytics Querying

Log Analytics provides a query syntax to quickly retrieve and consolidate data in the repository.



Query Language Syntax

A query's basic structure is a source table followed by a series of operators separated by the pipe character (|).

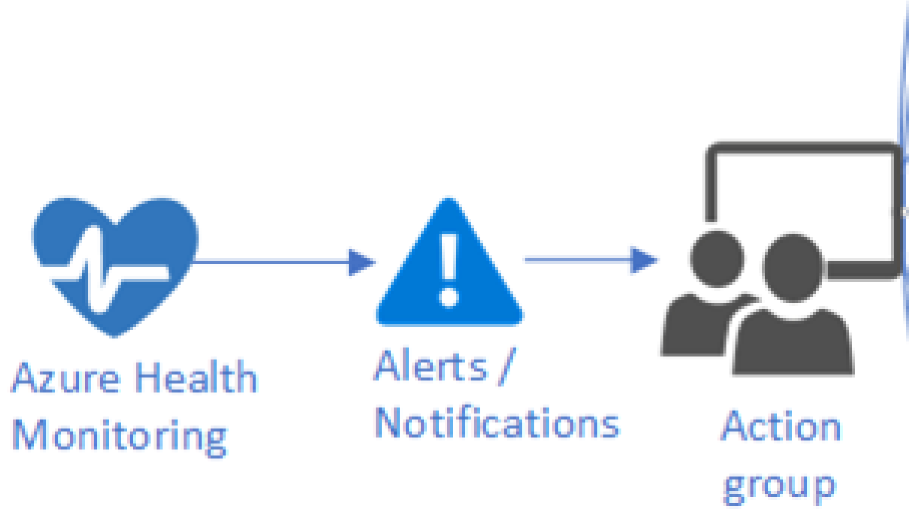


TECHNOLOGY

Choose a Mechanism for Event Routing and Escalation

Action Groups

It is a collection of notification preferences, defined by the owner of an Azure subscription.



Azure Monitor and Service Health alerts use action groups to notify users that an alert has been triggered.

Action Types

The seven action types in Azure portal are:

01 Automatic Runbook

02 Azure Function

03 Email Azure Resource Manager role

04 Email/SMS/Push/Voice

05 ITSM

06 Logic App

07 Webhook

Alerts

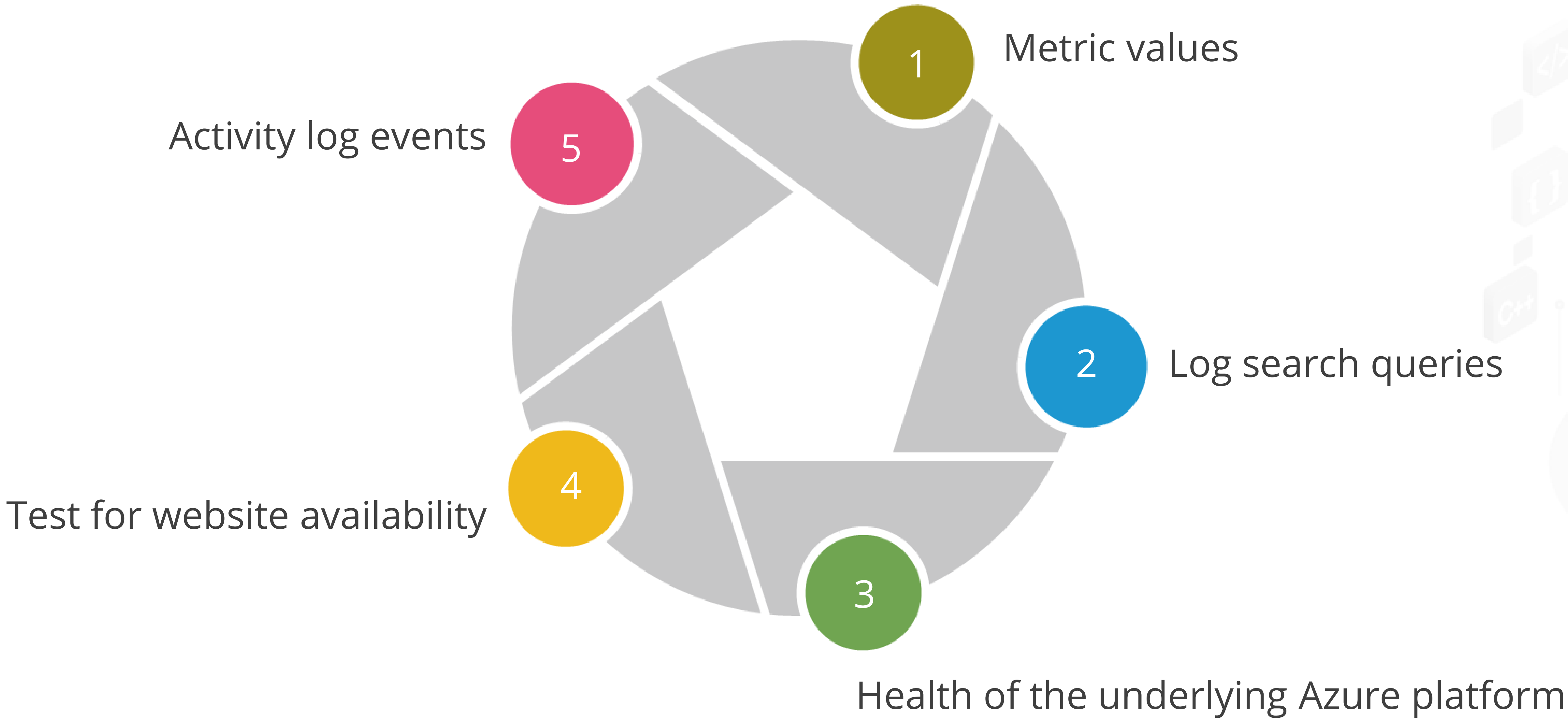
Alerts proactively notify a user when important conditions are found in the monitoring data.

Monitor alerts offer the following benefits:

- Better notification system
- A unified authoring experience
- Viewing of log analytics in the Azure portal
- Separation of fired alerts and alert rules
- Better workflow

Managing Alerts

Alerts can be set based on the following criteria:



Alert States

The key elements of alert states are:

New

The issue has just been detected and has not yet been reviewed.

Acknowledged

An administrator has reviewed the alert and started working on it.

Closed

The issue has been resolved. After an alert has been closed, the user can reopen it by changing it to another state.

Alert Rules

They are separated from alerts and the actions that are taken when an alert fires.

These are the key attributes of an alert rule:

- Target source
- Signal
- Criteria
- Alert name
- Alert description
- Security
- Action

Monitoring Azure

Duration: 10 Min.

Problem Statement:

You have been asked to create a proof of concept for monitoring the performance of virtual machines to ensure optimal operation and resource utilization

Outcome:

Users will have a proof of concept for monitoring the performance of Azure virtual machines.

Note: Refer to the demo document for detailed steps: 01_Monitoring_Azure

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to create Azure monitor are:

1. Deploy an Azure virtual machine
2. Create a Log Analytics workspace
3. Enable the Log Analytics virtual machine extension
4. Collect virtual machine event and performance data
5. View and query collected data

Key Takeaways

- Azure Monitoring collects and analyzes data to assess the application's performance, health, and availability.
- Azure Resource Health helps diagnose and get support for service problems that affect your Azure resources.
- Log Analytics helps collect and analyze data generated in the cloud and on-premise systems.
- An action group is a set of notification preferences set by the Azure subscriber.
- Alert rules are separate from alerts and actions taken when an alert fires.

Knowledge Check

Knowledge Check

1

Which of these has vulnerability management, vulnerability scanning, and incident management as its types?

A. Azure Update Management

B. Azure Infrastructure Monitoring

C. Azure Backup Services

D. Azure App Service

Knowledge Check 1

Which of these has vulnerability management, vulnerability scanning, and incident management as its types?

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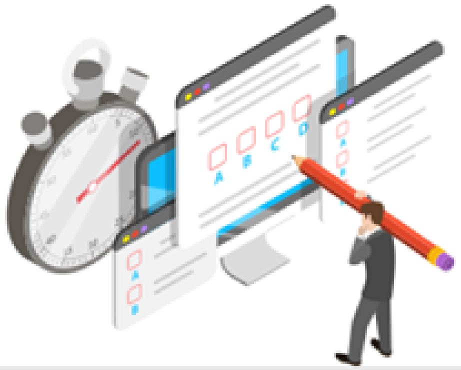


The correct answer is **B**

Vulnerability management, vulnerability scanning, and incident management are types of Azure infrastructure monitoring.

Knowledge Check 2

Which of the following is NOT a health status supported by Azure Resource Health?

A. Available

B. Degraded

C. Unknown

D. None of these

Knowledge Check
2

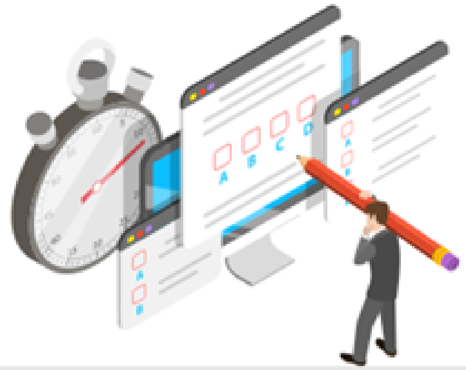
Which of the following is NOT a health status supported by Azure Resource Health?

A. Available

B. Degraded

C. Unknown

D. None of these



The correct answer is **D**

Azure Resource Health supports the following health statuses: available, unavailable, degraded, and unknown.

Knowledge Check 3

Which of these tools highlights over-provisioned services and recommends ways to lower costs?

A. Power BI Desktop

B. Azure Advisor Cost Analysis

C. Azure Advisor Recommendations

D. Consumption APIs

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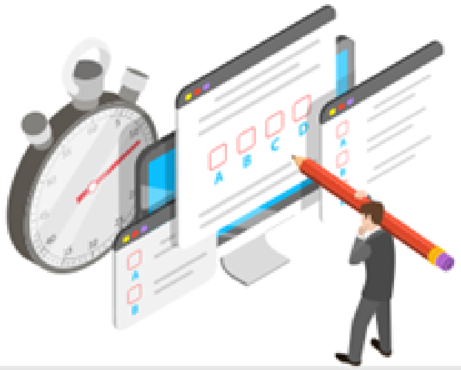


The correct answer is **C**

Azure Advisor recommendations highlight over-provisioned services and recommend ways to lower costs.

Knowledge Check

4

What terminology is used to describe a collection of notification preferences determined by the Azure subscription owner?

A. Azure monitor

B. Action group

C. Service health alert

D. Security

Knowledge Check 4

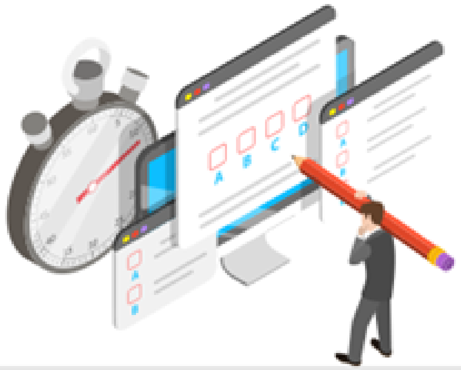
What terminology is used to describe a collection of notification preferences determined by the Azure subscription owner?

A. Azure monitor

B. Action group

C. Service health alert

D. Security



The correct answer is **B**

An action group is a collection of notification preferences that are determined by the owner of an Azure subscription.

Knowledge Check 5	Which of the following are the types of Azure infrastructure monitoring?

A. Vulnerability management

B. Vulnerability scanning

C. Incident management

D. All of these

Knowledge Check
5

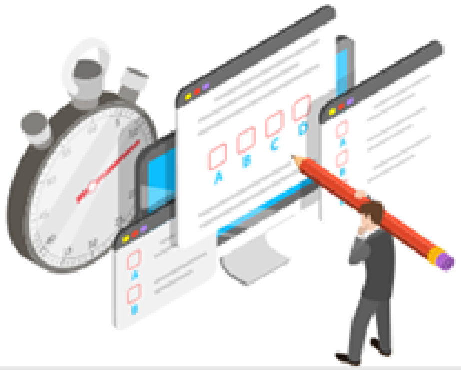
Which of the following are the types of Azure infrastructure monitoring?

A. Vulnerability management

B. Vulnerability scanning

C. Incident management

D. All of these



The correct answer is **D**

Vulnerability management, vulnerability scanning, and incident management are types of Azure infrastructure monitoring.

Case Study

Design for Logging and Monitoring

Scenario

You have taken a new position with Simplilearn Residences, which is very successful and is experiencing rapid growth. Simplilearn Residences is a building contractor for new homes and major home renovations and has become successful by providing quality buildings and offering newer integrated home technologies than their competitors. Currently, these technologies are provided and managed by separate subcontracting companies. The owner of Simplilearn Residences wants to begin offering these upgraded technology options in-house to provide better quality, support, and data on customer patterns and needs.

Design for Logging and Monitoring

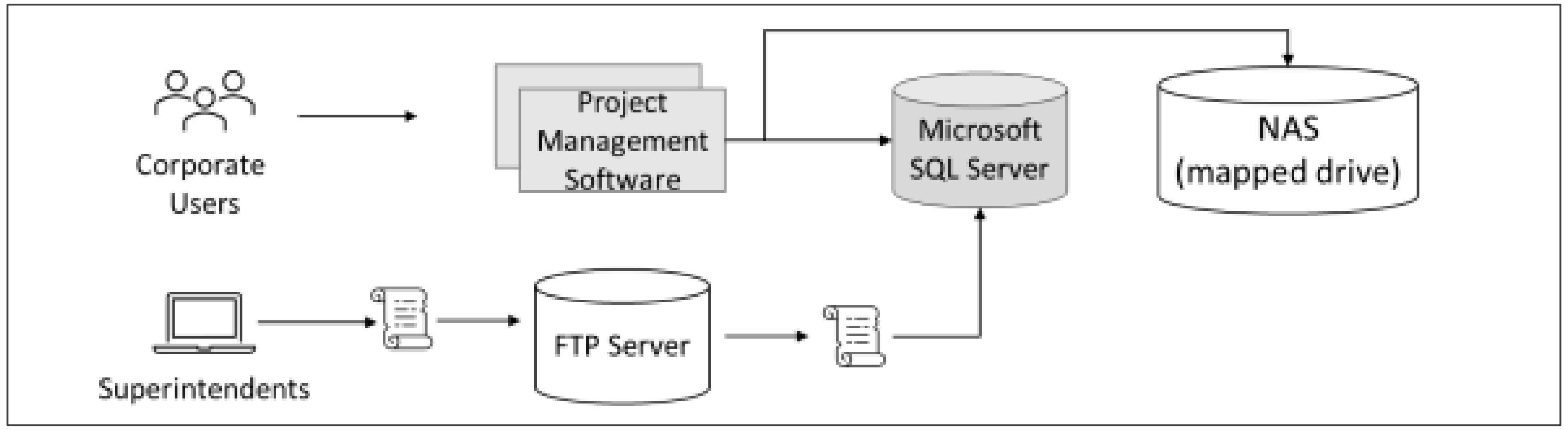
Scenario

Initially, the company wanted to offer HVAC (heating and cooling) control and monitoring, security system monitoring and alerts, and home automation. This will require a new website, a data storage solution, and a data ingestion solution. The company has seen tremendous growth over the past two years. The company is estimated to double its size over the next 12–18 months. With such rapid growth in the regional market, the company has no plans to expand outside the regional market.

Simplilearn Residences Headquarters runs a small data center in a single location. The data center hosts the company's Project Management (PM) software company.

Design for Logging and Monitoring

Scenario



- The PM software uses a third-party Windows application. The application runs on a 2-node Network Load Balancing (NLB) cluster with a single Microsoft SQL Server backend.
- Images and documents are stored on a mapped drive of the server, which exists on a dedicated NAS appliance.

Design for Logging and Monitoring

Scenario

- Corporate users, and office staff, use a web front end to enter data such as supply delivery schedules and change orders.
- Field superintendents use Windows laptops and tablets offline to continuously record building progress and other details. These changes, such as new work orders, are stored in a local change file. At the end of each day, superintendents return to the office to connect to the wireless network and run a small script to upload the change file to an FTP server. A second script is scheduled to run each night to process all the change files and enter their contents into the Project Management database (Microsoft SQL Server).

Design for Logging and Monitoring

Requirements

Project Management software:

- Migrate as many of the systems to a public cloud provider as possible.
- Replace the existing scripts to use a system more secure than FTP, as security concerns have arisen. Also, you have been asked to make sure that change files are processed as soon as they are uploaded.
- Increase the resilience of the project management database. While performance is not an issue, the company would like to avoid losing access to the database in case of a single hardware failure.

Design a solution for the Project Management software. Be prepared to explain why you chose each component of the design and how it meets the solution requirements

Design for Logging and Monitoring

Requirements

How are you incorporating the well-architected framework pillars to produce a high-quality, stable, and efficient cloud architecture?

Designing Solution for Logging and Monitoring

Duration: 10 min.

Project Agenda: To implement Azure Monitor alerts for the given scenario

Description: You have been given a project to design monitoring for a web application. You should be alerted whenever the number of HTTP 404 errors for the web app crosses a threshold of 5. You also need to ensure that appropriate stakeholders are notified about it

LESSON-END PROJECT

Design Governance

Duration: 20 min.

LESSON-END PROJECT

Perform the following:

1. Create a web app
2. Create an Azure alert